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Relationship between the G Tolerance, Physical Performance, and Cardiac Force Index in Male Aircrew: A Prospective Observational Study

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ABSTRACT

LAI, C.-Y., W.-W. HUANG, C.-L. KUO, W.-C. TSAI, C.-C. CHANG, and N.-J. CHANG. Relationship between the G Tolerance, Physical Performance, and Cardiac Force Index in Male Aircrew: A Prospective Observational Study. *Med. Sci. Sports Exerc.*, Vol. 57, No. 10, pp. 2227–2234, 2025. **Purpose:** Whether G tolerance is correlated physical performances and cardiac force index (CFI) in male aircrew. **Methods:** One month before the high G training, military flight cadets provided demographic data and details about their exercise habits. They wore BioPatch™ HP devices to track cardiac data, including walking CFI (WCFI), and during the 100-m sprint tests. Jump performance was assessed on the training day by measuring peak power output, peak power output normalized to total body mass, the product of force and time at the moment of the jump, and vertical jump height. Relaxed G tolerance and straining G tolerance were evaluated using a gradual onset rate. Logistic regression was used to analyze the relationships between G tolerance, physical performance, and CFI. **Results:** There were 53 male trainees classified in the low tolerance group and 29 ones in the high tolerance group. The finish time for the sprint in the high tolerance group averaged 14.1 ± 1.0 s, which was shorter than the 14.4 ± 1.1 s recorded for the low tolerance group. A higher WCFI (>0.15 vs ≤0.15) before the sprint was consistently associated with enhanced G tolerance performance, with an adjusted odds ratio (aOR) of 3.72 (95% confidence interval, 1.10–12.60). Conversely, the sprint completion time was negatively associated with G tolerance (adjusted odds ratio, 0.38; 95% confidence interval, 0.15–0.95). **Conclusions:** Our findings indicate that a higher WCFI and shorter sprint times prior to the high G training are positively correlated with the G tolerance. Aircrew training centers may integrate sprint and jump tests alongside cardiovascular monitoring to classify cadets into high or low G tolerance groups for further personalized training regimens. **Key Words:** ANTI-G STRAINING MANEUVER, COUNTERMOVEMENT JUMP, G-FORCE, G-INDUCED LOSS OF CONSCIOUSNESS, HIGH G TRAINING

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In the neutral standing posture, blood pressure decreases as the heart works to pump blood against gravity to the brain. On Earth, this phenomenon is influenced by the vertical gravity (known as the G-force), which affects the body axis and leads to the orthostatic pressure changes alongside the variations in the cardiac parameters (1–3). An increase in the G-force significantly reduces the blood supply to the brain. In response to the stress caused by the reduced blood flow detected by the baroreceptors, the autonomic nervous system becomes activated (4,5). Consequently, we can frequently observe or measure cardiovascular responses that alter the stroke volume, cardiac output (CO), heart rate (HR), and blood vessel resistance in order to maintain an adequate blood flow (6,7).

The G tolerance among military aircrew typically ranges from 4.5 to 6 G in a relaxed state (8). However, in the flight environment, the G-force in agile fighter jets can quickly escalate to levels of 7 to 9 times that of gravity. This poses a

significant risk of the G-induced loss of consciousness (GLOC), which can threaten flight safety. Surveys have indicated that approximately 10%–20% of military pilots have experienced GLOC at some point during their flying careers (9–13).

The timely and correct execution of the anti-G straining maneuver (AGSM) is a well-established method for preventing GLOC events. The AGSM involves isometric contractions of the lower body muscles and specific changes in breathing regulation. Among them, the straining of leg muscles is a crucial element in preventing blood pooling (14,15). The mechanism involves a significant increase in CO due to enhanced venous return and the resulting change in preload. However, this eventually becomes the limiting factor for CO augmentation, as the rise in HR is delayed. The operation of AGSM is similar to strength training or weightlifting, using a significant amount of energy over a short duration. Good physical fitness has been shown to improve the G tolerance duration and decrease the levels of physical fatigue and exertion during the centrifuge training (16–18).

Multiple studies have indicated that anthropometric characteristics and physiological responses alone may not effectively predict the G tolerance (18,19). To address this, the novel cardiac force index (CFI) was developed, which combines individual traits with cardiac responses. The CFI is calculated by multiplying body weight by activity level and dividing the result by the HR. The activity level is defined as the vector magnitude of three-axis (x, y, z) acceleration induced by body movement, measured in G, and detected by the triaxial accelerometer embedded in the wearable device. This value also reflects the intensity of physical exertion (20). Originally designed to prevent adverse events during military running tests, the CFI was later found to correlate with running performance (21–23). Because both high G training and running tests are high-intensity exercises that impose significant cardiovascular demands, we introduced the CFI as a tool to assess G tolerance. Our previous research sought to investigate the relationship between the CFI and G tolerance limits. Findings from various studies consistently revealed that the CFI at the relaxed walking stage (WCFI) measured before the high G training is positively correlated with the G tolerance (24,25).

As mentioned earlier, the key fundamental to proficiency in the AGSM is the strength of the lower body muscles, particularly the calves, hamstrings, quadriceps, and gluteal muscles. In addition, the AGSM operations resemble short, high-intensity physical exertion. Despite this, there has been limited research focused on assessing the physical condition related to the leg muscle performance (i.e., sprint and jump ability) and the cardiac regulation prior to the high G training. Therefore, the purpose of this current study was to assess physical fitness levels before high G training through both objective practical tests and subjective surveys on G tolerance, while concurrently monitoring cardiovascular responses. We hypothesized that the higher levels of physical fitness, particularly leg muscle performance (e.g., sprint and jump tests) and cardiovascular responses, are positively associated with G tolerance during high G training.

METHODS

Study Design

This research was conducted as a prospective observational study in 2023. This study was reviewed and approved by the Ethics Committee of the Institutional Review Board at Kaohsiung Armed Forces General Hospital, Kaohsiung City (Serial No. KAFGHIRB 112–029). Prior to the study, all participants were thoroughly informed about the research and signed consent forms, confirming their voluntary participation.

Participants

Inclusion criteria for this study were flight cadets from the Air Force Flight Training Command in Taiwan who had no prior experience with flight training and high G training. All participants had completed the annual health examination and met the standards for Flying Class I. None of the participants were using medications that could affect their HR, nor did they have any muscle strains or injuries. During the study period, military flight cadets lived in a group setting, followed the same daily schedule, and shared the same meals. Stressors related to G tolerance, such as fatigue and high physical exertion, were likely minimized or ignored to reduce variability.

Study Procedures

Flight cadets completed a questionnaire on exercise habits 1 month before high G training. Baseline tests included the 100-m sprint and the countermovement jump. G tolerance was assessed using a human centrifuge. A physician was present throughout the testing to ensure the safety of all participants. Illustrated in Figure 1, details of study protocol were divided into two sections as follows:

One-hundred-meter sprint test. After completing the questionnaire survey, the flight cadets participated in a 100-m sprint test on a standard 400-m track. Prior to the sprint, the BioPatch™ HP devices (Medtronic, Minneapolis, MN), which include the BioModule™ and its holder, were securely attached to their breastbones, just below the sternal notch (Fig. 2A). The cadets then engaged in a 10-min static warm-up to elevate their body temperature, enhance their joint range of motion, and prepare their muscles for exertion. This warm-up concluded with a 5-min session of light jogging.

The BioPatch™ HP monitoring devices were powered on after a 5-min resting period (Fig. 2B). Researchers verified that the green light indicator was flashing, signifying that data were being recorded normally. Following this, the flight cadets began a relaxed walk for 3 min (Fig. 2C). Before starting the walk and after completing it, they moved the devices up and down to label the data based on changes in vertical speed. When they were prepared at the starting line and heard the signal, they immediately sprinted to the finish line, covering a distance of 100 m (Fig. 2D). Each cadet was required to complete two sprints, separated by a 5-min interval. The time for each sprint was measured with an accuracy of 0.1 s.

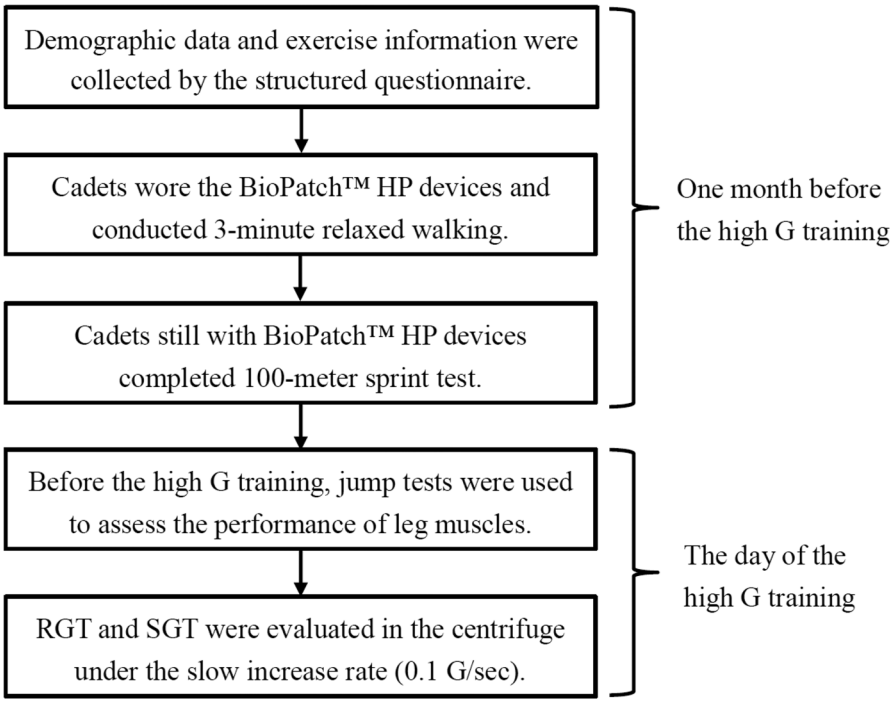


FIGURE 1—Study protocol.

Countermovement jump test. On the day of the high G training, the investigator provided an explanation and demonstration of the jump used to measure leg muscle performance. Prior to data collection, the flight cadets had the opportunity to practice the jump while wearing their flight suits and shoes. Each cadet completed three maximal effort attempts of the countermovement jump test using the SmartJump contact mat (FusionSport, Coopers Plains, Queensland, Australia). Before each jump, the force platform, which consisted of the contact mat and a wooden platform, was calibrated to zero, and the participants' weights were measured on the force platform.

During the test, subjects positioned themselves in an upright stance and then shifted to a semi-squat position, with their feet shoulder-width apart and elbows bent, hands resting on their waists. From this position, they jumped vertically into the air while keeping their hands firmly on their waists (26). Data collected from the jumps included the peak power output (PRO; in watts), peak power output normalized to total body mass (PROM; in watts per kilogram), impulse (the product of force and time at the moment of the jump, measured in newton-seconds (N·s)), and vertical jump height (VJH in centimeters).

G tolerance test. After the countermovement jump test, an instructor from the Aviation Physiology Research Laboratory provided a lecture briefing on the acceleration physiology and centrifuge profiles. Following this briefing, the subjects underwent the high G training so as to assess their relaxed G tolerance (RGT) and straining G tolerance (SGT). The human centrifuge (Latécoère, Toulouse, France) operated at a gradual onset rate of 0.1 $G \cdot s^{-1}$, powered by a hydraulic system. Initially, the flight cadets sat in a seat reclined at a 13° angle to begin the procedure. Inside the gondola, light bars were utilized to evaluate the extent of visual loss resulting from a decrease in brain-level blood pressure.

The RGT value was defined as the point at which the trainees experienced a complete loss of peripheral vision or a

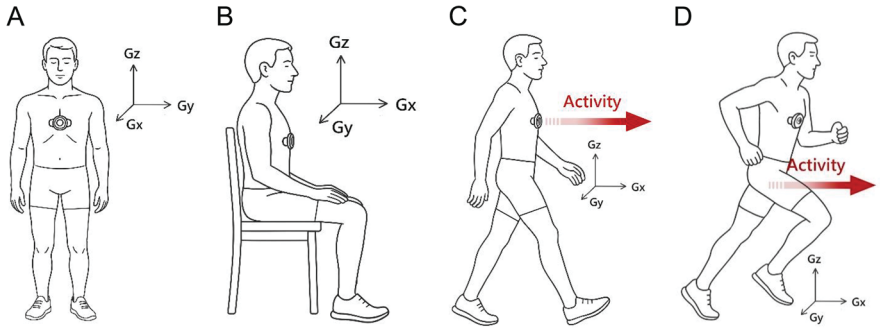


FIGURE 2—Measurement of HR responses and activity values. A, Attachment of BioPatch™ HP monitoring device. B, Resting stage for 5 min. C, Relaxed walk for 3 min. D, 100-m sprint test.

TABLE 1. Distribution of subject's characteristics and G tolerance.

Variables	Low Tolerance Group (n = 53)	High Tolerance Group (n = 29)	P Value
Age (yr)	23.5 ± 2.7	23.9 ± 3.4	0.780
Height (cm)	174.3 ± 5.5	171.2 ± 5.4	0.018
Weight (km)	70.7 ± 8.8	70.3 ± 8.8	0.668
BMI (kg·m^{-2})	23.2 ± 2.4	24.0 ± 2.5	0.229
RGT (G)	4.5 ± 0.6	5.9 ± 0.7	<0.001
≤5 G	42 (79.2%)	0 (0.0%)	
>5 G	11 (20.8%)	29 (100.0%)	
SGT (G)	7.8 ± 1.1	8.8 ± 0.3	<0.001
≤8 G	27 (50.9%)	0 (0.0%)	
>8 G	26 (49.1%)	29 (100.0%)	
AGSM effectiveness (G)	3.3 ± 1.1	2.9 ± 0.7	0.045
<3 G	20 (37.7%)	16 (55.2%)	0.198
≥3 G	33 (62.3%)	19 (44.8%)	

BMI, body mass index.

50% loss of central vision while in a relaxed state. Subsequently, they performed the AGSM to enhance the cardiovascular function in response to the G load. During this period, the centrifuge continued to gradually accelerate, with the SGT determined either when the subjects observed the same criteria of visual field loss reappearing or when the G-force reached the limit of 9 G. AGSM effectiveness was calculated as the difference between the RGT and SGT values.

Data Processing and Operation Definition

During the stages of the relaxed walking and sprint testing, the BioPatch™ HP monitoring device was used to measure a wide range of parameters. After completing the tests, we placed the device into its cradle and downloaded the targeted file labeled "Summary." According to the user manual, reliable data were identified by the HR confidence and system confidence levels greater than 20% (20). Signals with a sampling frequency of 1 s were retrieved based on the methodologies established in prior research (24,25).

The CFI was computed for each second using the formula: body weight × activity level/HR. The average value of the WCFI was derived from the data collected during the last 30 s of relaxed walking (25,26). Each subject performed the running tests twice, and we selected the data from the shorter completion time. For the 100-m sprint, we analyzed the two time intervals—1–5 s and 6–10 s—to calculate the mean of the sprint CFI (SCFI). In addition, during the three vertical jumps, we evaluated the performance of the highest jump by measuring the height, PRO, PROM, and impulse.

Sundblad et al. (27) divided participants into high and low G tolerance groups based on their RGT, facilitating a comparison of vasoconstrictor reserve responses between these two groups. Meanwhile, in the United States Air Force study, fighter pilots were more likely to pass the profile for sustaining 9 G for 15 s when their RGT and SGT were above 5 G and 8 G, respectively, under the gradual onset rate profile (18). As a result, flight cadets were categorized into a high tolerance group if they met the both aforementioned RGT and SGT thresholds; those who did not were classified as belonging to the low tolerance group.

Statistical Analysis

In the univariate analysis, we detailed the averages, standard deviations, counts, and proportions based on the types of variables examined. Independent variables related to the characteristics, exercise habits, cardiac responses, and physical performance were compared between the high and low tolerance groups.

Factors with P values less than 0.2, identified through univariate analysis, were included in the final multivariate model. A multiple logistic regression using the enter method was performed to examine the associations between G tolerance, physical fitness, and cardiac responses. The analysis was performed using the SPSS 27.0 software (IBM, Armonk, NY), and a significance level was set at a two-tailed P value of less than 0.05.

RESULTS

A total of 91 male flight cadets participated in this study. However, nine participants were excluded: seven due to poor data quality from the monitoring device and two for not completing the sprint test. Among the remaining 82 males, 53 were classified as the low tolerance group, whereas 29 (35.3%) achieved the high G tolerance.

As summarized in Table 1, there were no significant differences in age, weight, or body mass index between the low and high tolerance groups. However, the mean height was significantly shorter in the high tolerance group, measuring 171.2 ± 5.4 cm. In addition, the average values of the RGT and SGT in the high tolerance group were significantly greater than those in the low tolerance group (RGT: 5.9 ± 0.7 vs 4.5 ± 0.6, $P < 0.001$; SGT: 8.8 ± 0.3 vs 7.8 ± 1.1, $P < 0.001$). There was no significant difference in the proportions of the AGSM effectiveness below 3 G.

The high G tolerance group reported a slightly higher prevalence of aerobic and resistance training habits and longer durations of both activities compared with the low tolerance group. However, no significant differences were observed between the groups in terms of exercise habits or the focus on lower body resistance training (Table 2).

The mean WCFI before the sprint was 0.18 ± 0.11 kg·(G·min^{-1}) for high tolerance trainees and 0.15 ± 0.07 kg·(G·min^{-1}) for low tolerance trainees (P = 0.103). The average SCFI values during

TABLE 2. Habits and volumes of different exercises.

Variables	Low Tolerance Group (n = 53)	High Tolerance Group (n = 29)	P Value
Habit of aerobic exercise (%)	50 (94.3%)	29 (100.0%)	0.549
Duration of recent 3-mo aerobic exercise (min·wk^{-1})	97.9 ± 94.7	172.8 ± 192.6	0.095
Habit of resistance training (%)	39 (73.6%)	24 (82.8%)	0.504
Duration of recent 3-mo resistance training (min·wk^{-1})	106.5 ± 97.8	145.2 ± 146.5	0.570
Recent 3-mo resistance training for lower body muscles (%)	39 (73.6%)	22 (75.9%)	1.000

TABLE 3. Changes of cardiac response and performance of physical fitness.

Variables	Low Tolerance Group (n = 53)	High Tolerance Group (n = 29)	P Value
WCFI before the sprint (kg·(G·min^{-1}))	0.15 ± 0.07	0.18 ± 0.11	0.103
1- to 5-s SCFI (kg·(G·min^{-1}))	1.1 ± 0.3	1.0 ± 0.3	0.950
6- to 10-s SCFI (kg·(G·min^{-1}))	1.0 ± 0.3	0.9 ± 0.3	0.934
Sprint finish time (s)	14.4 ± 1.1	14.1 ± 1.0	0.121
Maximum VJH (cm)	34.7 ± 5.7	36.2 ± 4.8	0.187
PRO (W)	3254.0 ± 498.0	3327.4.9 ± 427.2	0.532
PROM (W·kg^{-1})	46.1 ± 5.6	47.5 ± 4.2	0.150
Impulse (N·s)	183.4 ± 25.9	186.5 ± 23.1	0.590

the 1- to 5-s and 6- to 10-s intervals did not show significant differences between the two groups. The finish time for the 100-m sprint in the high tolerance group was 14.1 ± 1.0 s, which was relatively shorter than the 14.4 ± 1.1 s recorded for the low tolerance group. Similar trends were observed in other physical fitness tests, including maximum VJH, RPO, PROM, and impulse (Table 3). We further observed the mild to moderate correlation between the cardiac response and physical fitness performance in these parameters (Table 4)

Multivariate logistic regression using the enter method was conducted to examine the cardiac responses and physical fitness factors associated with G tolerance. The analysis showed that high G tolerance was significantly positively associated with age (adjusted odds ratio (aOR), 1.28; 95% confidence interval (CI), 1.04–1.57) and with the WCFI (>0.15 vs ≤0.15: aOR, 3.72; 95% CI, 1.10–12.60). Conversely, G tolerance was significantly inversely associated with the 100-m sprint completion time (aOR, 0.38; 95% CI, 0.15–0.95). Among anthropometric variables, height was identified as a negative predictor of G tolerance (aOR, 0.82; 95% CI, 0.73–0.93) (Table 5).

DISCUSSION

Our team conducted this study to assess to integrate physical fitness and the CFI values into physical performance outcomes in a human centrifuge setting. This approach offers the advantage of potentially monitoring G tolerance through a simple running test or jump test combined with WCFI in everyday settings. Furthermore, these metrics could serve as valuable benchmarks for assessing the impact of physical exercise regimens.

In this study, selection bias may be a concern due to the young average age of participants, approximately 24 yr. In Taiwan, this reflects the Air Force flight cadet recruitment criteria, which require candidates to be between 21 and 27 yr old. According to the flight training schedule, individuals must complete and pass high G training at this age to qualify as fighter pilot candidates. The country invests substantial resources in military flight training; thus, ensuring early identification of G tolerance capacity is critical. Although this study focuses on a young cohort, its strength lies in utilizing an objective method to assess G tolerance at an early stage. By identifying individuals with low G tolerance, professionals can design personalized strength and physical training programs, ultimately safeguarding valuable flight personnel, improving cost-effectiveness, and enhancing future flight safety.

The CFI consists of three components: HR, activity level, and body weight. HR responses and body weight have been shown to correlate with G tolerance in military pilots (18,28,29). Moreover, HR activation is influenced by the type and intensity of physical activity, which also reflects varying activity levels. This connection serves as the primary rationale for applying the CFI to assess G tolerance. Previous studies have utilized wearable monitoring devices equipped with integrated three-axis accelerometers to collect HR and activity level data during different physical activities assisting for the calculation of the CFI.

In 2021, Chiang et al. (24) collected resting and walking HR and activity data prior to the centrifuge training. They calculated the CFI at different stages, reporting an average WCFI of 0.15 kg·(G·min^{-1}). Similarly, in 2023, Kuo et al. (25) found the mean value of the WCFI to be 0.14 kg·(G·min^{-1}). Our study demonstrated a comparable WCFI value prior to the sprint test. The participants in all three studies were flight cadets who met the stringent medical standards required during their annual health checkups, contributing to the consistency in physiological and demographic data across studies. In addition, our team ensured a consistent walking speed and pace during data collection by providing verbal reminders to participants, which likely minimized fluctuations in walking activity. This methodological consistency may explain the similar CFI levels observed across these studies, mitigating concerns about discrepancies in walking activity as a source of variation in CFI measurements.

TABLE 4. Correlation matrix between the cardiac response and performance of physical fitness.

Variables	RGT	SGT	WCFI	Sprint Finish Time	Maximum VJH	PRO	PROM	Impulse
RGT	1							
SGT	0.503**	1						
WCFI	0.096	–0.032	1					
Sprint finish time	–0.234*	0.099	0.019	1				
Maximum VJH	0.065	–0.072	–0.047	–0.532**	1			
PRO	0.049	–0.094	0.265*	–0.376**	0.543**	1		
PROM	0.083	–0.045	–0.011	–0.549**	0.969**	0.540**	1	
Impulse	0.033	–0.092	0.272*	–0.297**	0.391**	0.980**	0.372**	1

*P < 0.05.
**P < 0.01.

TABLE 5. Physical fitness related to G tolerance in the multivariate model.

Variables	Groups	β ± SE	aOR*	95% CI	P Value
Age (yr)		0.24 ± 0.11	1.28	1.04–1.57	0.021
Height (cm)		–0.19 ± 0.06	0.82	0.73–0.93	0.002
WCFI before the sprint (kg·(G·min^{-1}))	>0.15 vs ≤0.15	1.31 ± 0.62	3.72	1.10–12.60	0.035
Duration of recent 3-mo aerobic exercise (min·wk^{-1})		0.01 ± 0.01	1.00	0.99–1.01	0.109
Peak height (cm)		0.02 ± 0.06	1.04	0.90–1.16	0.708
Sprint finish time (s)		–0.98 ± 0.47	0.38	0.15–0.95	0.038

Consistent with previous findings, our results also confirmed that the CFI measured on the ground had a positive correlation with the G tolerance. Unlike earlier studies that assessed WCFI on the day of high G training (24,25), our research collected the WCFI data of the participants followed by a sprint test. This highlights the potential of WCFI as a valuable indicator for monitoring G tolerance levels over time in military pilots.

Regarding Table 2, although no variable reached statistical significance due to variability, consistent with previous finding, we observed that the high G tolerance group reported longer durations of aerobic training and a greater prevalence of regular resistance training habits. Meanwhile, the review study has shown that strength training, particularly exercises emphasizing isometric contractions and anaerobic power, is associated with improved G tolerance (16). Specifically, a study involving Polish Air Force cadets demonstrated that enhanced speed-strength abilities of the lower extremities correlated positively with G tolerance, suggesting that targeted strength training could be particularly beneficial for pilots (30). In addition, a moderate focus on aerobic fitness may aid in the rapid recovery from repeated G-force exposures and contribute to body composition changes that are favorable for high G flight (8,16).

This lack of correlation may be due to the cadets' focused physical fitness routines tailored specifically for G stress, resulting in a high prevalence of self-directed exercise habits. Being well-informed about the requirements and schedule for high G training, the cadets are likely to prepare effectively from the start. Moreover, previous studies have shown that G tolerance is influenced by multiple factors, underscoring the complexity of this relationship (17–19). Furthermore, reliance on the questionnaire is feasible but may have introduced bias into the results (e.g., the actual $\dot{V}O_{2max}$ was not measured).

Running and jump performances were assessed to evaluate the participants' physical fitness, particularly focusing on lower body strength. The 100-m run was selected as it closely mimics the duration and physical exertion required during a high G test. In this study, the high G tolerance group completed the 100-m sprint in 14.1 s, whereas the low G tolerance group took 14.4 s. Although the difference in sprint finish time (0.3 s) was not statistically significant, it represents a tangible and likely meaningful improvement in race performance, which could have practical implications for athletic achievement. We further evaluated its practical significance using the smallest worthwhile change and magnitude-based inference methods, which are well established in sports performance research (31–33). In sports science, statistically significant differences are not always practically meaningful, and conversely, nonsignificant findings can still represent important improvements (34). Magnitude-based inference analysis indicated a ~36% probability of a practically beneficial effect, with only ~17% risk of harm, reinforcing the notion that even small sprint improvements may be meaningful, particularly in high G tolerance contexts.

In this study, a moderate correlation between the indicators of the jump and running performances was found. In addition to the rapid and explosive muscle power output, subjects also needed to exhibit better cardiovascular responses and respiratory capacity to sustain their peak speed throughout the run. This suggests that the influence of the sprint running time on the G tolerance may be more significant than that of the jump performance. During the running process, we also calculated the changes in SCFI values every 5 s. Notably, the high tolerance group experienced a lower increase in HR during the 6- to 10-s interval of running (high vs low: 16.8 ± 11.7 vs 22.6 ± 12.1; $P = 0.036$).

Consistent with previous findings, a taller stature continued to be identified as an inverse factor affecting the G tolerance in this current study (18,35). To effectively prepare for the G stress, the significance of physical conditioning is frequently emphasized for military pilots. Despite this, no significant differences in exercise habits and volumes were observed among participants. However, subjective questionnaire data revealed considerable variability in the duration of aerobic and resistance training. Data from Santos et al. (36) reference values for body composition differed among the athletes. Prokopczyk and Wochyński (37) reported notable changes in body composition following a specific training program designed for future flight readiness. Therefore, these findings highlight the potential benefits of introducing more accessible and noninvasive methods for measuring body composition to objectively evaluate the effectiveness of physical training regimens.

It is essential to consider potential confounding variables that could influence the outcomes. First, participants consisted exclusively of military flight cadets who had already met stringent physical and medical standards for pilot training. This restriction in variability may have limited the generalizability of the findings to broader populations with more diverse physical fitness levels. Second, the use of self-reported questionnaire data to assess exercise habits may have introduced recall bias or subjective inaccuracies. Objective measures, such as $\dot{V}O_{2max}$ testing or direct monitoring of training routines, could provide more reliable insights. Third, ensuring maximum effort during running and jump tests was challenging. Although cadets were encouraged to exert maximum effort during tests, variability in motivation and exertion levels could have influenced performance metrics, potentially leading to an underestimation of performance metrics. Fourth, CFI data during the jump test were unable to capture

due to the technological limitations (e.g., brief reaction time). Fifth, the study included only male flight cadets, which limits the applicability of findings to female pilots or other populations. Six, future longitudinal studies to test whether targeted interventions, such as increasing aerobic exercise, can improve sprint times and subsequently enhance G tolerance.

CONCLUSIONS

Our study provides additional insights into the predictors of G tolerance in military trainees by integrating sprint time and CFI as key variables. A novel aspect of this research lies in quantifying the relationship between WCFI and sprint performance within a controlled military setting, offering practical implications for optimizing training regimens. Prior to high G training, we successfully measured WCFI and physical performance metrics among military flight cadets. Our findings indicate that higher WCFI values and shorter sprint times are positively correlated with greater G tolerance. These results support the incorporation of the 100-m sprint test and WCFI measurements into routine fitness assessments for military pilots. As simple and noninvasive tools, these tests could facilitate early identification of individuals with lower G-force tolerance. In addition, implementing longitudinal studies within training programs would enable the evaluation of how sustained physical improvements impact G tolerance over time, thereby refining and enhancing training strategies.

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